

Python Introduction

For those who can already code!

Thank You

- Credits: Frank Van Puyvelde
Original repo: https://github.com/franklbvp/python_intro
This repo is forked in September 2025 for some updates
- Geert Jan Bex
 - <https://github.com/gjbex/training-material/tree/master/Python>
 - <https://creativecommons.org/licenses/by/4.0/deed.ast#>
- (updated) Slides + demo-files
 - https://github.com/moravveji/python_intro_training

What will be covered in this training?

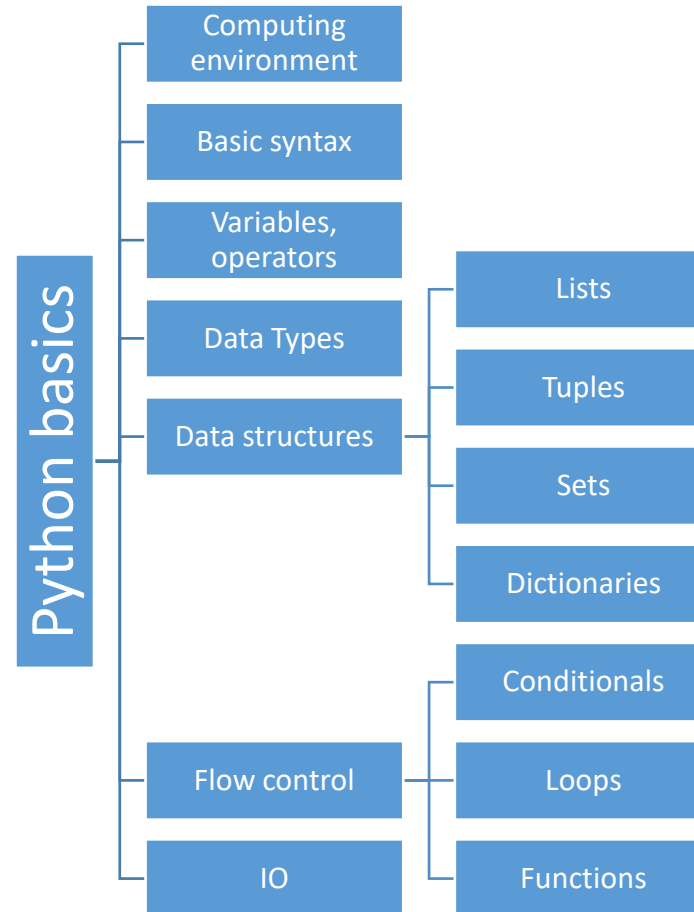
Programming with Python: ingredients

- **Python Interpreter:** the actual program that reads and executes the Python code you write.
- **Text Editor or Integrated Development Environment (IDE):** where you write Python code.
- Basic Understanding of **Python Syntax:** includes knowledge of Python's data types, control flow structures, and functions.
- **Knowledge of Python Libraries:** Python libraries are collections of functions and methods that allow to perform many actions without writing your own code.
- **Problem Solving Skills:** Programming is all about solving problems. Break down complex problems into smaller, more manageable parts.
- **Patience and Persistence:** you will encounter errors and bugs, and things won't work as expected. Debug your code and keep trying until you get it right.
- **Just do it:** the best way to learn Python programming is by doing.

Bonus topics

- Hands on exercises
- Python programming using AI
- Advanced VSC Python trainings

Python basics roadmap



Scope

- Getting to know the very basics:
 - How to run basic code in Python.
 - What are the basic programming blocks in Python?
- These sessions won't teach you how to program, how to find algorithms, ... that's beyond the scope
 - No idea about programming, consider completing
 - CodeAcademy
<https://www.codecademy.com/catalog/language/python>
 - LearnPython
<https://www.learnpython.org/>
- *Taken from GJ Bex*

Contents

- Computing Environment
 - Command line
 - Visual Studio Code (or one may choose Spyder)
 - Notebook
 - *Slides: Python_intro-userEnvironments*

Contents

- Installing Python
 - Slides: *02-Python_intro-Setup*
- Basic syntax
 - Slides: *03-Python_intro-syntax*
- Variables and Operators
 - Slides: *04-Python_intro-variables_operators*
- Built-in atomic Data types
 - Slides: *05-Python_intro-builtin_datatypes*
- Data structures (compound data types)
 - Lists
 - Tuples
 - Sets
 - Dictionaries
 - Slides: *06-Python_intro-datastructures*

Contents

- Flow control
 - Basic flow control commands
 - *Slides: 07-Python_intro-control_flow*
 - Functions
 - *Slides: 08-Python_intro-functions*
- IO
 - *Slides: 09-Python_intro-io*
- [Bonus] Programming with AI
 - *Slides: 10-Python_intro-programming-with-AI*
- [Bonus] Python on the HPC platform
- [Bonus] Python learning tracks

Using the demo files

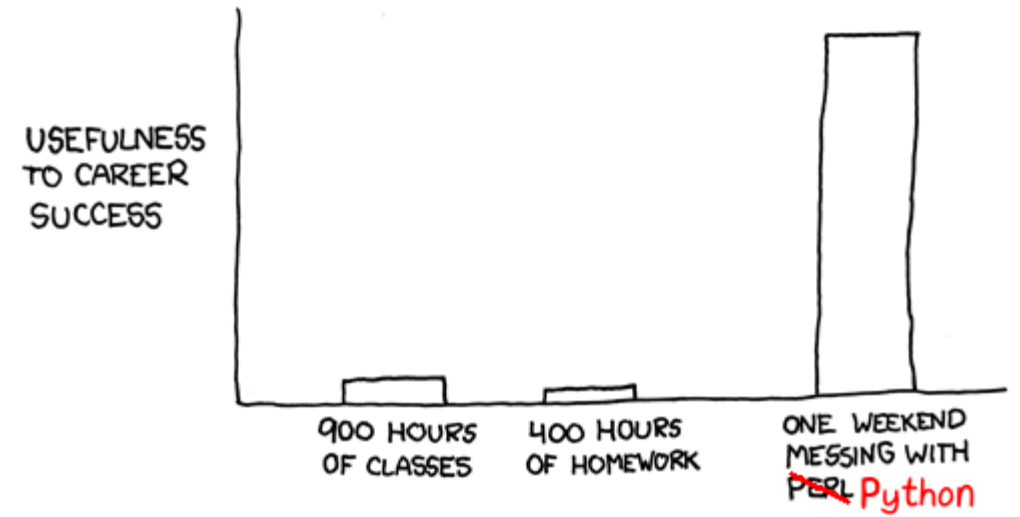
- Using the demo notebooks
- Download the zip file
- Unzip the file a local disk (Windows ex. in c:\Temp)
- Run notebook
 - Start powershell python environment
 - 'cd' to the folder with ipynb files
 - `jupyter lab`
- `py` source files
 - 'cd' to the folder with py files
 - Start editor / IDE
 - Run from command line

How popular is Python (late 2025)?

- [Tiobe index](#)
- September 2025
- General trend (not HPC languages)
- Also [IEEESpectrum](#)
- Also [GitHub 2.0](#)

 TIOBE < the software quality company >		About us	Knowledge	News	Coding Standards	TIOBE Index	Contact	🔍
		Products ▾	Quality Models ▾	Markets ▾	Schedule a demo			
Sep 2025	Sep 2024	Change	Programming Language		Ratings	Change		
1	1			Python	25.98%	+5.81%		
2	2			C++	8.80%	-1.94%		
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4	3	▼		Java	8.35%	-1.09%		
5	5			C#	6.38%	+0.30%		
6	6			JavaScript	3.22%	-0.70%		
7	7			Visual Basic	2.84%	+0.14%		
8	8			Go	2.32%	-0.03%		

University activities

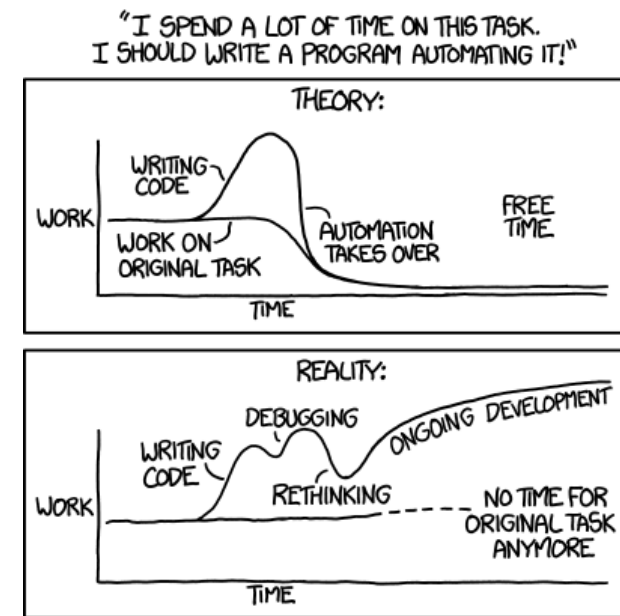


https://fabienmaussion.info/scientific_programming/img/00_messing_python.png

Programming 101

Why programming?

- Programming is an integral part of research.
- Programming occurs on different levels:
 - Write small scripts,
 - Write complete projects,
 - Need a good understanding of what a software package does.
- All programming languages offer to a certain extend the same building blocks.
 - Understand the basic building blocks.
 - Decompose your problem to fit those blocks.



xkcd 1319: Automation

Why write programs for research?

- Scripted research can be tested and reproduced
- Programs are a rigorous way of describing data analysis for other researchers, as well as for computers.
- Code can be shared
- Code is much more easy to understand for a non-author than spreadsheets

Programming?

- It may be (almost) impossible to solve a problem by executing commands at the command prompt.
- What is needed? A **sequence of precise instructions** that, once performed, will complete a **specific task**.
- Computer programs can't do that many things, they can:
 - Assign values to variables (memory locations).
 - Make decisions based on comparisons.
 - Repeat a sequence of instructions over and over.
 - Call subprograms.

Programming language

- There are many programming languages, with changing popularity
- Check the Tiobe Index: <https://www.tiobe.com/tiobe-index/>
- Consider:
 - it is suited to the problem at hand?
 - is there an active community?
 - is it any good for the job market?

Key concepts in programming

- All programming languages use the same basic concepts.
- Check Isaac Computer Science:
https://isaaccomputerscience.org/topics/programming_concepts?examBoard=all&stage=all
- Instructions / Basic Syntax
- Data Types
 - Classification of the type of data being stored or manipulated within a program.
 - Data types are important because they determine the operations that can be performed on the data.
- Variables
 - Named container, held by the computer in a memory location.
 - Has a unique identifier (name) that refers to a value.
- Input / Output

Key concepts in programming

- Operators
 - Arithmetic
 - Comparison
 - Logical
- Sequence:
statements are written one after another, will be executed one statement at a time in the order that the statements are written in.
- Selection:
execute lines of code only if a certain condition is met.
- Iteration (loop):
repeat a group of statements .
- Subprogram (function):
is a named sequence of statements, can be repeatedly “called” from different places in the program

And there will be errors...

- Syntax error
 - A mistake against the language rules
 - Program will not run and will return an error message
- Runtime error
 - Usually due to some missing variables, modules,...
- Semantic error
 - A mistake in the reasoning
 - Program is not executing as intended / expected

Python

A quickstart into the very basics

Get to know your Python environment

Thank You

- <https://github.com/gjbex/training-material/tree/master/Python>
- *Whirlwind Tour of Python* by Jake VanderPlas
<https://jakevdp.github.io/WhirlwindTourOfPython/>
- <https://www.cs.cornell.edu/courses/cs1110/2023fa/materials/python/>
- https://fabienmaussion.info/scientific_programming/welcome.html
- <https://justinbois.github.io/bootcamp/>

See also

- <https://fangohr.github.io/teaching/python/book.html>
- <https://patrickwalls.github.io/mathematicalpython/>
- https://sites.duke.edu/compsci_101l_001_sp24/
- <https://github.com/parrt/msan501>
- <https://docs.python-guide.org/intro/learning/>

Tutorials

- <https://www.python.org/about/gettingstarted/>
- <https://realpython.com/>
- <https://realpython.com/matlab-vs-python/>
- <https://www.learnpython.org/>
- Cheat sheets
- <https://www.datacamp.com/community/data-science-cheatsheets>

Python: setting the scene

get comfortable within the Python universe

What is Python?

- From www.python.org: "Python is an interpreted, object-oriented, high-level programming language with dynamic semantics. Its high-level built in data structures, combined with dynamic typing and dynamic binding, make it very attractive for Rapid Application Development, as well as for use as a scripting or glue language to connect existing components together. Python's simple, easy to learn syntax emphasizes readability and therefore reduces the cost of program maintenance."
- Python is a general purpose programming language used for a huge variety of purposes. It's user community is growing rapidly!
(<https://stackoverflow.blog/2017/09/06/incredible-growth-python/>)

What is Python?

- a **general purpose interpreted** programming language.
- a language that supports multiple approaches to software design, principally **structured** and **object-oriented** programming.
- provides **automatic memory management** and **garbage collection**
- **dynamically** typed.

Why Python?

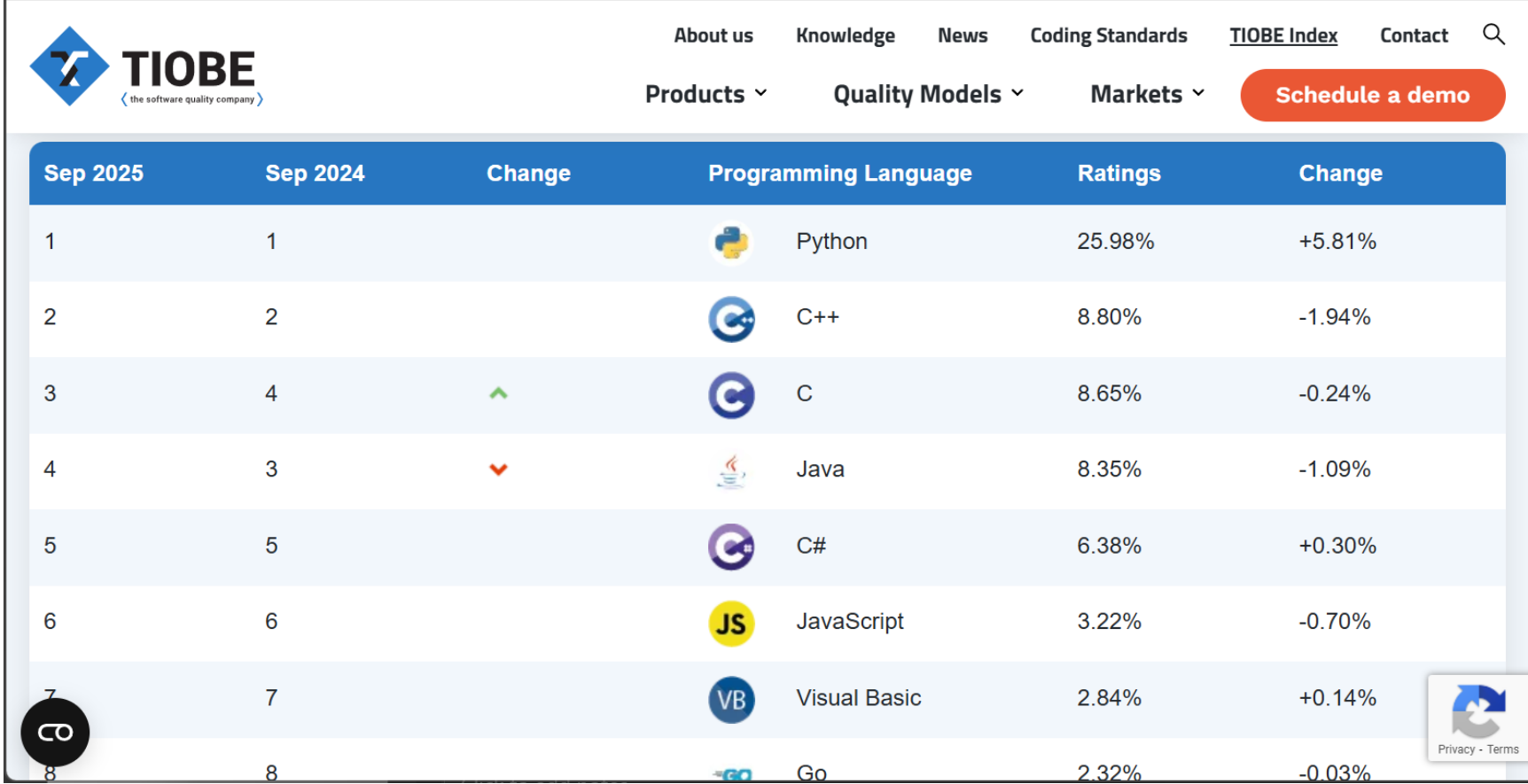
- Python is free
- Cross-platform (Windows, Mac, Linux)
- Has a vibrant user community (StackOverflow)
- Rich 3rd party packages, e.g. Conda, PyPI
- Python is quick to program in, easy exploration of modeling constructs, and fast analysis of data. (*explorative programming*)
- Python is popular in research, and has lots of libraries for science
 - extensive capabilities, documentation, and support
 - Access to advanced math, statistics, and database functions
- Python interfaces well with faster languages (C/C++, Fortran)

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What is Python used for?

<https://www.python.org/about/apps/>



The screenshot shows the TIOBE Index website. The header includes the TIOBE logo, navigation links (About us, Knowledge, News, Coding Standards, TIOBE Index, Contact), and a 'Schedule a demo' button. The main content is a table of the top 8 programming languages. The table has columns for rank in Sep 2025, rank in Sep 2024, change, programming language, ratings, and change in ratings. Python is at the top with a 25.98% rating and a +5.81% change. Other languages include C++, C, Java, C#, JavaScript, Visual Basic, and Go.

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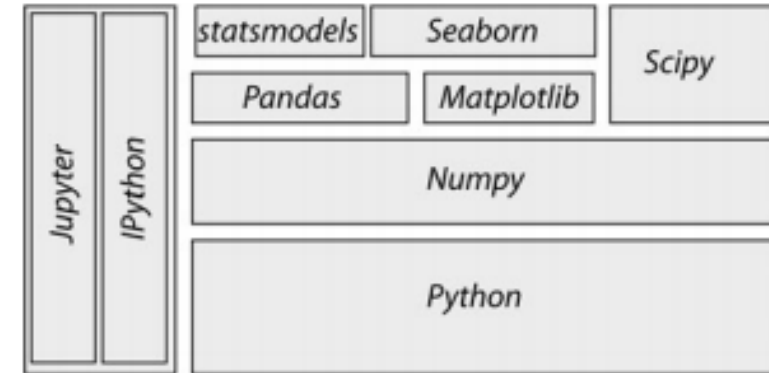
Python ecosystem

- Large and active ecosystem

- Core Python

- Standard libraries
 - third-party packages:

- *NumPy* for manipulation of homogeneous array-based data,
 - *Pandas* for manipulation of heterogeneous and labeled data,
 - *SciPy* for common scientific computing tasks,
 - *Matplotlib* for publication-quality visualizations,
 - *IPython* for interactive execution and sharing of code, etc. *Python versions*



Python versions

- Current 3.x
 - More clean than 2.x
 - Python 3.x introduced some backwards-incompatible changes to the language, so code written for 2.7 may not work under 3.x and vice versa.
 - Almost all Python libraries supported
- Version 2.7.x
 - Last of the 2.x releases
 - Many Python 3.x features have been retrofitted
 - All libraries support it

Note: in-application scripting
may be stuck at Python 2.7!

Python 2 countdown:
<https://pythonclock.org/>

- *Taken from GJ Bex*